UNIVERSITY OF DETROIT MERCY
COLLEGE OF ENGINEERING & SCIENCE

CSSE-3430 Data Structures & Algorithms
Syllabus – Fall 2026
The University of Detroit Mercy
In Partnership with Detroit Green Technology Institute (DGTI)

Course Description

Review of object-oriented principles, standard data structures, Big-O notation, NP-completeness, stacks, queues, generic data types, dynamic memory, recursion, linked lists, circular lists, doubly linked lists, trees, binary search trees, heaps, graphs, sorting algorithms, searching algorithms, object-oriented language implementation of data structures and algorithms.

Pre-requisites:

CSSE 1720 OR CSSE 1722 (Min Grade of "C", may not be taken concurrently)

Ability to program using the C++ programming language with an understanding of abstract data types

Lecturer:

Ma'en Saleh, Ph.D.

Email: salehms3@udmercy.edu

Office hours

Days	China-time	Notes
Tuesday	4.00-5.00 PM	Send me an email to arrange meetings outside the regular schedule
Wednesday	1.00-2.00 PM	
Wednesday	4.00-5.00 PM	

Teaching Assistants:

1. Chen Yonghui e-mail: hg_cyh@hbut.edu.cn
2. Tang Zehua email: 1055294671@qq.com

Class Meetings:

Format [modify as needed]: Online synchronous class using Blackboard collaborate. Attendance is mandatory.

Days	China-time	Notes
Monday	2.00-3.35 PM	Weeks 6-13 (7A-215)
Wednesday	2.00-3.35 PM	Weeks 6-13 (7A-215)
Thursday	3.55-5.30 PM	Weeks 6-13 (7A-215)

Course Objectives :

1. Demonstrate a familiarity with major algorithms and data structures.
2. Apply important algorithmic design paradigms and methods of analysis.
3. Synthesize efficient algorithms in common engineering design situations.
4. Use the appropriate combination of algorithms and data structure to write efficient programs.

Course Outcomes:

Upon completion of the course, students will be able to:

1. Understand the abstraction of object-oriented programming.
2. Manipulate dynamically allocated memory.
3. Implement basic data structures (for instance, stacks, queues, linked lists) using arrays.
4. Implement basic data structures (for instance, stacks, queues, linked lists) using pointers.
5. Distinguish among common time complexities.
6. Work in teams.
7. Implement and analyze searching and sorting algorithms.

Program Learning Outcomes:

Graduates of this program will be able to:

1. Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.
2. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.
3. Communicate effectively in a variety of professional contexts.
4. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.
6. Apply computer science theory and software development fundamentals to produce computing-based solutions.

List of Topics:

1. Introduction to data structures
2. Algorithm Analysis
   - Asymptotic Analysis
   - Searching: Sequential and Binary search
3. C++ Review and Templates
4. ADT, STL, Iterators, and Vector
5. List, Stack and Queue
6. Sorting Algorithms
   - STL sort, Insertion, Shell, Merge, and Quick sort
7. Trees:
   - Binary tree and Binary search tree

Recommended Textbook:

1. Mark Allen Weiss, **Data Structures and Algorithm Analysis in C++**, 4th edition, Pearson, 2014
   Mark Allen Weiss, **Data Structures and Algorithm Analysis in Java**, 3rd edition, Pearson, 2012

2. Clifford Shaffer, **Data Structures and Algorithm Analysis**, Edition 3.2 (C++ Version), 2013.
   Clifford Shaffer, **Data Structures and Algorithm Analysis**, Edition 3.2 (Java Version), 2013.

3. Michael T. Goodrich, Roberto Tamassia, and David M. Mount (2011), **Data Structures and Algorithms in C++**, Wiley, 6th Edition, ISBN: 978-0470383278

4. Frank Carrano and Timothy Henry (2017), **Data Abstractions & Problem Solving with C++**, Pearson, 7th edition, ISBN: 978-0137516759

5. Mark A. Weiss (2013), **Data Structures and Algorithm Analysis in C++**, Pearson, 4rd Edition**,** ISBN: 978-0133404180

Course Tentative Schedule

- **Week 1 (Oct. 5 – Oct. 11): Introduction to Data Structures**
  - **Topics:**
    - **Overview of data structures**
    - **Importance and applications**
    - **Abstract Data Types (conceptual introduction)**
    - **Basic data representation (arrays, pointers, references)**
    - **Complexity intuition: time vs. space trade-offs**
    - **Real-world examples of data structures**
  - **Lab 0: C++ Warm-Up & Environment Setup**
  - **COs: 1, 2**

- **Week 2 (Oct. 12 – Oct. 18): Algorithm Analysis**
  - **Topics:**
    - **Asymptotic analysis**
    - **Sequential search**
    - **Binary search**
  - **Lab 1: Time Complexity and Searching**
  - **COs: 1, 2, 6**

- **Week 3 (Oct. 19 – Oct. 25): C++ Review and Templates**
  - **Topics:**
    - **Essential C++ concepts for data structures**
    - **Function and class templates**
  - **Lab 2: Template Functions and Classes**

- **COs: 2, 6**

- **Week 4 (Oct. 26 – Nov. 1): Abstract Data Types (ADT) and STL Basics**
  - **Topics:**
    - **ADT principles**
    - **Introduction to STL containers**
    - **Iterators overview**
  - **Lab 3: Using STL Containers and Iterators**
  - **COs: 1, 2, 6**

- **Week 5 (Nov. 2 – Nov. 8): Midterm Exam + Vectors and Iterators in Depth**
  - **Topics:**
    - **Vector implementation**
    - **Vector traversal techniques**
  - **Lab 4: Implementing and Testing Vectors**
  - **Midterm Exam**
  - **COs: 1, 2, 3, 6**

- **Week 6 (Nov. 9 – Nov. 15): Lists**
  - **Topics:**
    - **Linked list implementation**
    - **Operations and applications**
  - **Lab 5: Singly and Doubly Linked Lists**
  - **COs: 1, 2, 6**

- **Week 7 (Nov. 16 – Nov. 22): Stacks and Queues**
  - **Topics:**
    - **Stack implementation and applications**
    - **Queue implementation and applications**
  - **Lab 6: Implementing Stacks and Queues**
  - **COs: 1, 2, 6**

- **Week 8 (Nov. 23 – Nov. 29): Sorting Algorithms**
  - **Topics:**
    - **STL sort**
    - **Insertion sort**
    - **Merge sort**
    - **Quick sort**
  - **Lab 7: Implementing and Comparing Sorting Algorithms**
  - **Final Exam (theory) + Final Exam (lab)**
  - **COs: 1, 2, 6**

Note: Assignments and quizzes will be announced on Blackboard.

DGTI Policies [REQUIRED]:

Cell Phones Policy:

The use of cell phones (including voice, text, and any other applications not required for the course) is **not allowed** during lectures and other class activities. Cell phones must be **turned off or left at home** so as not to disturb the other students in the class. The consequences of using a cell phone or a cell phone ringing during class can include one or more of the following:

- The instructor will answer the call and/or share the communication with the entire class;
- The owner of the cell phone will be immediately asked to leave the classroom and may not return during that class period. Any work missed cannot be made up;
- The entire class will be given an immediate quiz; and
- The use or ringing of a cell phone during an exam will result in the termination of the exam at that point for the owner of the phone

* If you receive an emergency phone call, leave the classroom or laboratory and take the phone call in the hallway or outdoors.

Academic Expectations and Courtesy:

Students are expected to adhere to the following academic expectations and courtesy behavior:

- Students are expected to always conduct themselves in a **professional manner**. This expectation extends to **all** course related activities including the use of email;
- Email communications should be **professionally written;**
- Students must be courteous to their fellow classmates, teaching assistant, and professor. Talking during lectures is **not permitted**. If you are disturbing others in class, the professor will require you to leave; and
- Academic Policy requires students to submit any final grade concerns no later than 5 days after students have received official final grades from the administration. Specific dates are mentioned in the Grade Distribution section of the Syllabus.

Help with Assignments:

The professor and teaching assistants (if available for this course) are willing to help with the assignments in the class. However, students must demonstrate that they have made a good-faith effort at solving the problem before consulting with the professor or teaching assistants.

When asking questions, be prepared to demonstrate the efforts that you have made in solving the problem and be prepared to discuss your plan of attack. Statements such as “I don‘t understand the problem, please show me how to do it” will likely **not** be answered. You will be asked about what you do understand about the problem. Questions should be brought to the professor or teaching assistants well before the due date: Neither the professor nor the teaching assistant are obligated to make extensive homework consulting time available the day before the assignment due date. The professor and teaching assistants will make an effort to answer email and discussion board inquiries within 24 hours, but this is **not** guaranteed.

Grading for the course will be based primarily on, but may vary according to courses:

- **Attendance & class participation** (absences may lead to failure)
    - Students are required to **attend and be punctual** at every session.

- Timely submission of **homework /Assignments** (due dates)
    - The failure of a disk, a computer, or other storage media is not an acceptable excuse for late or missing homework assignments
- **Quizzes**
    - The professor reserves the right to give both announced and unannounced quizzes
- **Exams/ tests** (midterm and final exam)
    - Exam ownership: Faculty owns the exam and copying is strictly prohibited. If copying and/or photography is used, the student will automatically fail the class. You will be able to view completed exams but may not keep them or photograph them.
    - Legitimate excuse (i.e., illness, death in the family, etc.), you must contact the professor within 24 hours of the examination date (with proof, such as a doctor's note). Otherwise, you will receive a zero for that exam.
- Independent **project** /term project (with written reports and oral presentations)

Academic Integrity:

As members of an academic community engaged in the pursuit of truth and with a special concern for values, University of Detroit Mercy students must conform to the highest standard of honesty and integrity in their academic work. The fundamental assumption under which the University operates is that **work submitted by a student is a product of their own efforts**. Among the most serious academic offensives is **plagiarism**, submitting the ideas or work of another source without acknowledgment or documentation. The consequences of plagiarism or any act of academic dishonesty may range from failure on a course to dismissal from the university.

The professor takes academic integrity **very seriously**, and violations will be reported to the appropriate College authorities.

- All work submitted for this course must be the sole the work of the student or the team of students working on a group assignment.
- Any work (words, ideas, data, code, etc.) that is taken from another source must be properly cited and documented.

Violations of Academic Integrity:

Violations of academic integrity includes, but is not limited to the following:

1) Plagiarism:
    (a) Copying ideas or taking exact wording from published sources without using quotation marks and indicating where the words came from.
    (b) Paraphrasing from sources without citation.
    (c) Copying someone else's work and submitting it as your own.
    (d) Buying a paper or having someone else write it for you.

2) Inappropriate Collaboration:
    (a) Working together on individual tale-home exams.
    (b) Collaborating with students outside of the course on individual take-home exams.
    (c) Using student's work from previous years.
    (d) Handing in assignments (typically electronic submissions) where only superficial changes have been made.

3) Cheating:

(a) Using materials not permitted by the instructor during exams or assignments.

(b) Inappropriate copying answers from other students (on exams or assignments).

(c) Altering graded exams or assignments and submitting them for regrading in order to get a better grade

4) Facilitating Academic Dishonesty:

(a) Allowing another student to copy an assignment that was given as individual work.

(b) Allowing another to copy exam answers.

(c) Taking exams or completing assignments for another student.

Copyrights:

All course materials students receive or have online access to **are protected by copyright laws**. In addition, the textbook and the professor-provided resources are likewise **protected**. Students may use course materials and make copies for their own use as needed, but unauthorized distribution and/or uploading of materials without the professor's express permission is strictly **prohibited**. Students who engage in the unauthorized distribution of copyrighted materials may be held in violation of the Student Code of Conduct, and/or liable under Federal and State laws. In addition, Copying, reposting, and submitting course materials (like completed essays, labs, homework, exams, quizzes, or other assignments ） to other websites or information repositories constitutes a violation of the Student Conduct policy and are **strictly prohibited**.

Recording of Class Sessions:

Video/audio recordings of class sessions (face-to-face or online) may be recorded (by the professor). Recordings will be shared via platforms with access limited to other members of the class.

Students may **not** record the professor without consent during lectures, laboratories, office hours, appointments, or discussions. The use of audio and/or video recording equipment or the making of audio recordings, video recordings, or transcriptions is not permitted. The use of all recording devices, including tape recorders, video cameras, still cameras, electronic voice recorders, computers, cell phones, tablets, and any other recording devices is not permitted.

Grading Policy

Programming Lab Assignments	20%
Final lab Exam	10%
Quizzes & Written Assignments	10%
Attendance	5%
Midterm Exam	25%
Final Exam	30%

Programming Assignments (In-Class Labs):

At the end of each topic, students will work individually during lab sessions on a programming assignment. Both the quality of the submitted work and the student's progress and participation during class will be evaluated. Completing and submitting these assignments is essential to passing the course, as they directly support your understanding of the material and prepare you for exams.

- Submission Requirements: Each student must submit their own work, including both the source code and the program output.
    - All assignments must be submitted on time.
    - Late submissions will be accepted for up to 3 days, with a maximum grade of 75%.
- Grading Criteria:
    - Correctness: 60%
    - Program Design: 15%
    - Style & Documentation: 10%
    - In-Class Progress & Participation: 15%

NOTE: Students with questions related to their course grade or achievement for FALL 2026 must contact instructors by Monday, November 30th, 2026 at 1:00 PM China-Time. Any grade inquiry beyond specified date might not be addressed.

Grading Scale:

Percentage	Letter Grade
93 – 100%	A
90 - 92%	A^-
87 - 89%	B^+
83 – 86%	B
80 – 82%	B^-
77 – 79%	C^+
73 – 76%	C
70 – 72%	C^-
65 – 69%	D^+
60 – 64%	D
< 60%	F

Important Dates:

Note: *The instructor reserves the right to change the outline to accommodate student pace in understanding of the subject matter.*

Follow Blackboard Weekly activities as posted on BB class site